T, contradictory constraints arise. Since any witness of satisfiability has to assign some value to that node, we infer that it cannot be T. Thus, we may permanently assign mark F to that node. For this DAG, such an optimization does not seem to help. No test of an unmarked node detects a shared mark or a shared contradiction. Our cubic SAT solver fails for this DAG.

1.7 Exercises

Exercises 1.1

1. Use $\neg$, $\rightarrow$, $\wedge$ and $\vee$ to express the following declarative sentences in propositional logic; in each case state what your respective propositional atoms p, q, etc. mean:
   * (a) If the sun shines today, then it won't shine tomorrow.
   (b) Robert was jealous of Yvonne, or he was not in a good mood.
   (c) If the barometer falls, then either it will rain or it will snow.
   * (d) If a request occurs, then either it will eventually be acknowledged, or the requesting process won't ever be able to make progress.
   (e) Cancer will not be cured unless its cause is determined and a new drug for cancer is found.
   (f) If interest rates go up, share prices go down.
   (g) If Smith has installed central heating, then he has sold his car or he has not paid his mortgage.
   * (h) Today it will rain or shine, but not both.
   * (i) If Dick met Jane yesterday, they had a cup of coffee together, or they took a walk in the park.
   (j) No shoes, no shirt, no service.
   (k) My sister wants a black and white cat.
2. The formulas of propositional logic below implicitly assume the binding priorities of the logical connectives put forward in Convention 1.3. Make sure that you fully understand those conventions by reinserting as many brackets as possible. For example, given $p \wedge q \rightarrow r$, change it to $(p \wedge q) \rightarrow r$ since $\wedge$ binds more tightly than $\rightarrow$.
   * (a) $\neg p \wedge q \rightarrow r$
   (b) $(p \rightarrow q) \wedge \neg(r \vee p \rightarrow q)$
   * (c) $(p \rightarrow q) \rightarrow (r \rightarrow s \vee t)$
   (d) $p \vee (\neg q \rightarrow p \wedge r)$
   * (e) $p \vee q \rightarrow \neg p \wedge r$
   (f) $p \vee p \rightarrow \neg q$
   * (g) Why is the expression $p \vee q \wedge r$ problematic?

Exercises 1.2

1. Prove the validity of the following sequents:
   (a) $(p \wedge q) \wedge r, s \wedge t \vdash q \wedge s$

(b) $p \wedge q \vdash q \wedge p$
* (c) $(p \wedge q) \wedge r \vdash p \wedge (q \wedge r)$
(d) $p \to (p \to q), p \vdash q$
* (e) $q \to (p \to r), \neg r, q \vdash \neg p$
* (f) $\vdash (p \wedge q) \to p$
(g) $p \vdash q \to (p \wedge q)$
* (h) $p \vdash (p \to q) \to q$
* (i) $(p \to r) \wedge (q \to r) \vdash p \wedge q \to r$
* (j) $q \to r \vdash (p \to q) \to (p \to r)$
(k) $p \to (q \to r), p \to q \vdash p \to r$
* (l) $p \to q, r \to s \vdash p \vee r \to q \vee s$
(m) $p \vee q \vdash r \to (p \vee q) \wedge r$
* (n) $(p \vee (q \to p)) \wedge q \vdash p$
* (o) $p \to q,\ r \to s \vdash p \wedge r \to q \wedge s$
(p) $p \to q \vdash ((p \wedge q) \to p) \wedge (p \to (p \wedge q))$
(q) $\vdash q \to (p \to (p \to (q \to p)))$
* (r) $p \to q \wedge r \vdash (p \to q) \wedge (p \to r)$
(s) $(p \to q) \wedge (p \to r) \vdash p \to q \wedge r$
(t) $\vdash (p \to q) \to ((r \to s) \to (p \wedge r \to q \wedge s))$; here you might be able to 'recycle' and augment a proof from a previous exercise.
(u) $p \to q \vdash \neg q \to \neg p$
* (v) $p \vee (p \wedge q) \vdash p$
(w) $r, p \to (r \to q) \vdash p \to (q \wedge r)$
* (x) $p \to (q \vee r), q \to s, r \to s \vdash p \to s$
* (y) $(p \wedge q) \vee (p \wedge r) \vdash p \wedge (q \vee r)$.

2. For the sequents below, show which ones are valid and which ones aren't:
* (a) $\neg p \to \neg q \vdash q \to p$
* (b) $\neg p \vee \neg q \vdash \neg(p \wedge q)$
* (c) $\neg p, p \vee q \vdash q$
* (d) $p \vee q, \neg q \vee r \vdash p \vee r$
* (e) $p \to (q \vee r), \neg q, \neg r \vdash \neg p$ without using the MT rule
* (f) $\neg p \wedge \neg q \vdash \neg(p \vee q)$
* (g) $p \wedge \neg p \vdash \neg(r \to q) \wedge (r \to q)$
(h) $p \to q, s \to t \vdash p \vee s \to q \wedge t$
* (i) $\neg(\neg p \vee q) \vdash p$.

3. Prove the validity of the sequents below:
(a) $\neg p \to p \vdash p$
(b) $\neg p \vdash p \to q$
(c) $p \vee q, \neg q \vdash p$
* (d) $\vdash \neg p \to (p \to (p \to q))$
(e) $\neg(p \to q) \vdash q \to p$
(f) $p \to q \vdash \neg p \vee q$
(g) $\vdash \neg p \vee q \to (p \to q)$

(h) $p \to (q \vee r),\ \neg q,\ \neg r \vdash \neg p$
(i) $(c \wedge n) \to t,\ h \wedge \neg s,\ h \wedge \neg(s \vee c) \to p \vdash (n \wedge \neg t) \to p$
(j) the two sequents implict in (1.2) on page 20
(k) $q \vdash (p \wedge q) \vee (\neg p \wedge q)$ using LEM
(l) $\neg(p \wedge q) \vdash \neg p \vee \neg q$
(m) $p \wedge q \to r \vdash (p \to r) \vee (q \to r)$
* (n) $p \wedge q \vdash \neg(\neg p \vee \neg q)$
(o) $\neg(\neg p \vee \neg q) \vdash p \wedge q$
(p) $p \to q \vdash \neg p \vee q$ possibly without using LEM?
* (q) $\vdash (p \to q) \vee (q \to r)$ using LEM
(r) $p \to q,\ \neg p \to r,\ \neg q \to \neg r \vdash q$
(s) $p \to q,\ r \to \neg t,\ q \to r \vdash p \to \neg t$
(t) $(p \to q) \to r,\ s \to \neg p,\ t,\ \neg s \wedge t \to q \vdash r$
(u) $(s \to p) \vee (t \to q) \vdash (s \to q) \vee (t \to p)$
(v) $(p \wedge q) \to r,\ r \to s,\ q \wedge \neg s \vdash \neg p$.

4. Explain why intuitionistic logicians also reject the proof rule PBC.
5. Prove the following theorems of propositional logic:
* (a) $((p \to q) \to q) \to ((q \to p) \to p)$
(b) Given a proof for the sequent of the previous item, do you now have a quick argument for $((q \to p) \to p) \to ((p \to q) \to q)$?
(c) $((p \to q) \wedge (q \to p)) \to ((p \vee q) \to (p \wedge q))$
* (d) $(p \to q) \to ((\neg p \to q) \to q)$.
6. Natural deduction is not the only possible formal framework for proofs in propositional logic. As an abbreviation, we write Γ to denote any finite sequence of formulas $\phi_1, \phi_2, \ldots, \phi_n$ $(n \geq 0)$. Thus, any sequent may be written as $\Gamma \vdash \psi$ for an appropriate, possibly empty, Γ. In this exercise we propose a different notion of proof, which states rules for transforming valid sequents into valid sequents. For example, if we have already a proof for the sequent $\Gamma, \phi \vdash \psi$, then we obtain a proof of the sequent $\Gamma \vdash \phi \to \psi$ by augmenting this very proof with one application of the rule $\to$i. The new approach expresses this as an inference rule between sequents:

$$\frac{\Gamma, \phi \vdash \psi}{\Gamma \vdash \phi \to \psi} \to\text{i}.$$

The rule 'assumption' is written as

$$\frac{}{\phi \vdash \phi}\ \text{assumption}$$

i.e. the premise is empty. Such rules are called axioms.
(a) Express all remaining proof rules of Figure 1.2 in such a form. (Hint: some of your rules may have more than one premise.)
(b) Explain why proofs of $\Gamma \vdash \psi$ in this new system have a tree-like structure with $\Gamma \vdash \psi$ as root.
(c) Prove $p \vee (p \wedge q) \vdash p$ in your new proof system.

7. Show that $\sqrt{2}$ cannot be a rational number. Proceed by proof by contradiction: assume that $\sqrt{2}$ is a fraction k/l with integers k and $l \neq 0$. On squaring both sides we get $2 = k^2/l^2$, or equivalently $2l^2 = k^2$. We may assume that any common 2 factors of k and l have been cancelled. Can you now argue that $2l^2$ has a different number of 2 factors from k^2? Why would that be a contradiction and to what?
8. There is an alternative approach to treating negation. One could simply ban the operator $\neg$ from propositional logic and think of $\phi \to \bot$ as 'being' $\neg\phi$. Naturally, such a logic cannot rely on the natural deduction rules for negation. Which of the rules $\neg$i, $\neg$e, $\neg\neg$e and $\neg\neg$i can you simulate with the remaining proof rules by letting $\neg\phi$ be $\phi \to \bot$?
9. Let us introduce a new connective $\phi \leftrightarrow \psi$ which should abbreviate $(\phi \to \psi) \wedge (\psi \to \phi)$. Design introduction and elimination rules for $\leftrightarrow$ and show that they are derived rules if $\phi \leftrightarrow \psi$ is interpreted as $(\phi \to \psi) \wedge (\psi \to \phi)$.

Exercises 1.3

In order to facilitate reading these exercises we assume below the usual conventions about binding priorities agreed upon in Convention 1.3.

1. Given the following formulas, draw their corresponding parse tree:
   (a) p
   * (b) $p \wedge q$
   (c) $p \wedge \neg q \to \neg p$
   * (d) $p \wedge (\neg q \to \neg p)$
   (e) $p \to (\neg q \vee (q \to p))$
   * (f) $\neg((\neg q \wedge (p \to r)) \wedge (r \to q))$
   (g) $\neg p \vee (p \to q)$
   (h) $(p \wedge q) \to (\neg r \vee (q \to r))$
   (i) $((s \vee (\neg p)) \to (\neg p))$
   (j) $(s \vee ((\neg p) \to (\neg p)))$
   (k) $(((s \to (r \vee l)) \vee ((\neg q) \wedge r)) \to ((\neg(p \to s)) \to r))$
   (l) $(p \to q) \wedge (\neg r \to (q \vee (\neg p \wedge r)))$.
2. For each formula below, list all its subformulas:
   * (a) $p \to (\neg p \vee (\neg\neg q \to (p \wedge q)))$
   (b) $(s \to r \vee l) \vee (\neg q \wedge r) \to (\neg(p \to s) \to r)$
   (c) $(p \to q) \wedge (\neg r \to (q \vee (\neg p \wedge r)))$.
3. Draw the parse tree of a formula ϕ of propositional logic which is
   * (a) a negation of an implication
   (b) a disjunction whose disjuncts are both conjunctions
   * (c) a conjunction of conjunctions.
4. For each formula below, draw its parse tree and list all subformulas:
   * (a) $\neg(s \to (\neg(p \to (q \vee \neg s))))$
   (b) $((p \to \neg q) \vee (p \wedge r) \to s) \vee \neg r$.

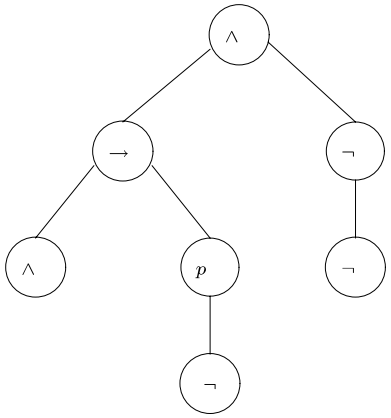


Figure 1.21. A tree that represents an ill-formed formula.

* 5. For the parse tree in Figure 1.22 find the logical formula it represents.

6. For the trees below, find their linear representations and check whether they correspond to well-formed formulas:
   (a) the tree in Figure 1.10 on page 44
   (b) the tree in Figure 1.23.

* 7. Draw a parse tree that represents an ill-formed formula such that
   (a) one can extend it by adding one or several subtrees to obtain a tree that represents a well-formed formula;
   (b) it is inherently ill-formed; i.e. any extension of it could not correspond to a well-formed formula.

8. Determine, by trying to draw parse trees, which of the following formulas are well-formed:
   (a) $p \wedge \neg(p \vee \neg q) \rightarrow (r \rightarrow s)$
   (b) $p \wedge \neg(p \vee q \wedge s) \rightarrow (r \rightarrow s)$
   (c) $p \wedge \neg(p \vee \wedge s) \rightarrow (r \rightarrow s)$.
   Among the ill-formed formulas above which ones, and in how many ways, could you 'fix' by the insertion of brackets only?

Exercises 1.4

* 1. Construct the truth table for $\neg p \vee q$ and verify that it coincides with the one for $p \rightarrow q$. (By 'coincide' we mean that the respective columns of `T` and `F` values are the same.)

2. Compute the complete truth table of the formula
   * (a) $((p \rightarrow q) \rightarrow p) \rightarrow p$
   (b) represented by the parse tree in Figure 1.3 on page 34

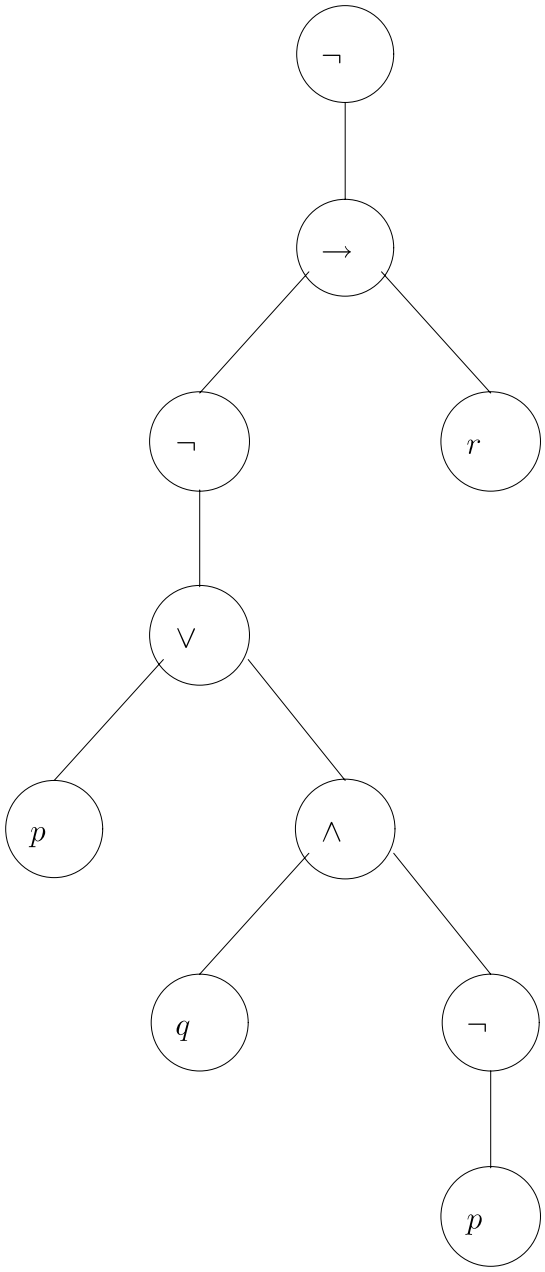


Figure 1.22. A parse tree of a negated implication.

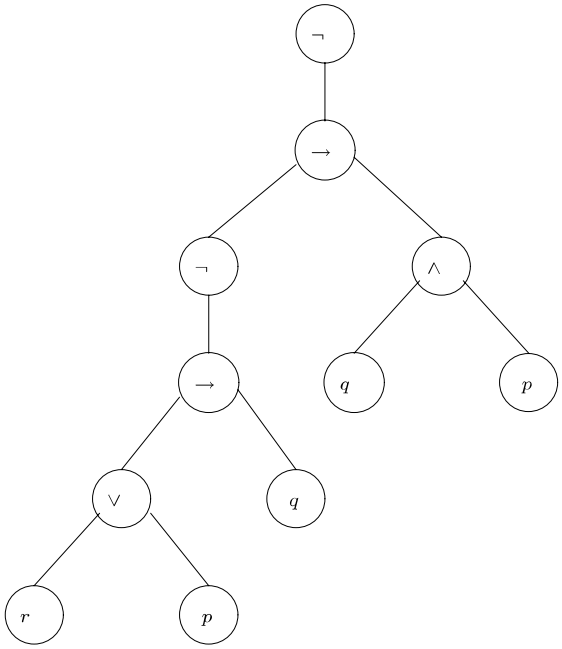


Figure 1.23. Another parse tree of a negated implication.

* (c) $p \vee (\neg(q \wedge (r \to q)))$
(d) $(p \wedge q) \to (p \vee q)$
(e) $((p \to \neg q) \to \neg p) \to q$
(f) $(p \to q) \vee (p \to \neg q)$
(g) $((p \to q) \to p) \to p$
(h) $((p \vee q) \to r) \to ((p \to r) \vee (q \to r))$
(i) $(p \to q) \to (\neg p \to \neg q)$.

3. Given a valuation and a parsetree of a formula, compute the truth value of the formula for that valuation (as done in a bottom-up fashion in Figure 1.7 on page 40) with the parse tree in
* (a) Figure 1.10 on page 44 and the valuation in which q and r evaluate to `T` and p to `F`;
(b) Figure 1.4 on page 36 and the valuation in which q evaluates to `T` and p and r evaluate to `F`;
(c) Figure 1.23 where we let p be `T`, q be `F` and r be `T`; and
(d) Figure 1.23 where we let p be `F`, q be `T` and r be `F`.

4. Compute the truth value on the formula's parse tree, or specify the corresponding line of a truth table where
* (a) p evaluates to `F`, q to `T` and the formula is $p \to (\neg q \vee (q \to p))$
* (b) the formula is $\neg((\neg q \wedge (p \to r)) \wedge (r \to q))$, p evaluates to `F`, q to `T` and r evaluates to `T`.

* 5. A formula is valid iff it computes `T` for all its valuations; it is satisfiable iff it computes `T` for at least one of its valuations. Is the formula of the parse tree in Figure 1.10 on page 44 valid? Is it satisfiable?

6. Let $*$ be a new logical connective such that $p * q$ does not hold iff p and q are either both false or both true.
   (a) Write down the truth table for $p * q$.
   (b) Write down the truth table for $(p * p) * (q * q)$.
   (c) Does the table in (b) coincide with a table in Figure 1.6 (page 38)? If so, which one?
   (d) Do you know $*$ already as a logic gate in circuit design? If so, what is it called?

7. These exercises let you practice proofs using mathematical induction. Make sure that you state your base case and inductive step clearly. You should also indicate where you apply the induction hypothesis.
   (a) Prove that

$$(2 \cdot 1 - 1) + (2 \cdot 2 - 1) + (2 \cdot 3 - 1) + \cdots + (2 \cdot n - 1) = n^2$$

   by mathematical induction on $n \geq 1$.
   (b) Let k and l be natural numbers. We say that k is divisible by l if there exists a natural number p such that $k = p \cdot l$. For example, 15 is divisible by 3 because $15 = 5 \cdot 3$. Use mathematical induction to show that $11^n - 4^n$ is divisible by 7 for all natural numbers $n \geq 1$.
* (c) Use mathematical induction to show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6}$$

   for all natural numbers $n \geq 1$.
* (d) Prove that $2^n \geq n + 12$ for all natural numbers $n \geq 4$. Here the base case is $n = 4$. Is the statement true for any $n < 4$?
   (e) Suppose a post office sells only 2¢ and 3¢ stamps. Show that any postage of 2¢, or over, can be paid for using only these stamps. Hint: use mathematical induction on n, where n¢ is the postage. In the inductive step consider two possibilities: first, n¢ can be paid for using only 2¢ stamps. Second, paying n¢ requires the use of at least one 3¢ stamp.
   (f) Prove that for every prefix of a well-formed propositional logic formula the number of left brackets is greater or equal to the number of right brackets.

* 8. The Fibonacci numbers are most useful in modelling the growth of populations. We define them by $F_1 \stackrel{\text{def}}{=} 1$, $F_2 \stackrel{\text{def}}{=} 1$ and $F_{n+1} \stackrel{\text{def}}{=} F_n + F_{n-1}$ for all $n \geq 2$. So $F_3 \stackrel{\text{def}}{=} F_1 + F_2 = 1 + 1 = 2$ etc. Show the assertion 'F_{3n} is even.' by mathematical induction on $n \geq 1$. Note that this assertion is saying that the sequence $F_3, F_6, F_9, \ldots$ consists of even numbers only.

9. Consider the function rank, defined by

$$\begin{aligned}\mathsf{rank}(p) &\stackrel{\text{def}}{=} 1\\ \mathsf{rank}(\neg\phi) &\stackrel{\text{def}}{=} 1 + \mathsf{rank}(\phi)\\ \mathsf{rank}(\phi \circ \psi) &\stackrel{\text{def}}{=} 1 + \max(\mathsf{rank}(\phi), \mathsf{rank}(\psi))\end{aligned}$$

where p is any atom, $\circ \in \{\rightarrow, \vee, \wedge\}$ and $\max(n, m)$ is n if $n \geq m$ and m otherwise. Recall the concept of the height of a formula (Definition 1.32 on page 44). Use mathematical induction on the height of ϕ to show that $\mathsf{rank}(\phi)$ is nothing but the height of ϕ for all formulas ϕ of propositional logic.

* 10. Here is an example of why we need to secure the base case for mathematical induction. Consider the assertion

'The number $n^2 + 5n + 1$ is even for all $n \geq 1$.'

(a) Prove the inductive step of that assertion.
(b) Show that the base case fails to hold.
(c) Conclude that the assertion is false.
(d) Use mathematical induction to show that $n^2 + 5n + 1$ is odd for all $n \geq 1$.

11. For the soundness proof of Theorem 1.35 on page 46,
(a) explain why we could not use mathematical induction but had to resort to course-of-values induction;
(b) give justifications for all inferences that were annotated with 'why?' and
(c) complete the case analysis ranging over the final proof rule applied; inspect the summary of natural deduction rules in Figure 1.2 on page 27 to see which cases are still missing. Do you need to include derived rules?

12. Show that the following sequents are not valid by finding a valuation in which the truth values of the formulas to the left of $\vdash$ are $\mathtt{T}$ and the truth value of the formula to the right of $\vdash$ is $\mathtt{F}$.
(a) $\neg p \vee (q \rightarrow p) \vdash \neg p \wedge q$
(b) $\neg r \rightarrow (p \vee q), r \wedge \neg q \vdash r \rightarrow q$
* (c) $p \rightarrow (q \rightarrow r) \vdash p \rightarrow (r \rightarrow q)$
(d) $\neg p, p \vee q \vdash \neg q$
(e) $p \rightarrow (\neg q \vee r), \neg r \vdash \neg q \rightarrow \neg p$.

13. For each of the following invalid sequents, give examples of natural language declarative sentences for the atoms p, q and r such that the premises are true, but the conclusion false.
* (a) $p \vee q \vdash p \wedge q$
* (b) $\neg p \rightarrow \neg q \vdash \neg q \rightarrow \neg p$
(c) $p \rightarrow q \vdash p \vee q$
(d) $p \rightarrow (q \vee r) \vdash (p \rightarrow q) \wedge (p \rightarrow r)$.

14. Find a formula of propositional logic ϕ which contains only the atoms p, q and r and which is true only when p and q are false, or when $\neg q \wedge (p \vee r)$ is true.

15. Use mathematical induction on n to prove the theorem $((\phi_1 \wedge (\phi_2 \wedge (\cdots \wedge \phi_n)\ldots) \rightarrow \psi) \rightarrow (\phi_1 \rightarrow (\phi_2 \rightarrow (\ldots(\phi_n \rightarrow \psi)\ldots))))$.
16. Prove the validity of the following sequents needed to secure the completeness result for propositional logic:
    (a) $\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \rightarrow \phi_2)$
    (b) $\neg\phi_1 \wedge \neg\phi_2 \vdash \phi_1 \rightarrow \phi_2$
    (c) $\neg\phi_1 \wedge \phi_2 \vdash \phi_1 \rightarrow \phi_2$
    (d) $\phi_1 \wedge \phi_2 \vdash \phi_1 \rightarrow \phi_2$
    (e) $\neg\phi_1 \wedge \phi_2 \vdash \neg(\phi_1 \wedge \phi_2)$
    (f) $\neg\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \wedge \phi_2)$
    (g) $\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \wedge \phi_2)$
    (h) $\neg\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \vee \phi_2)$
    (i) $\phi_1 \wedge \phi_2 \vdash \phi_1 \vee \phi_2$
    (j) $\neg\phi_1 \wedge \phi_2 \vdash \phi_1 \vee \phi_2$
    (k) $\phi_1 \wedge \neg\phi_2 \vdash \phi_1 \vee \phi_2$.
17. Does $\vDash \phi$ hold for the ϕ below? Please justify your answer.
    (a) $(p \rightarrow q) \vee (q \rightarrow r)$
    * (b) $((q \rightarrow (p \vee (q \rightarrow p))) \vee \neg(p \rightarrow q)) \rightarrow p$.

Exercises 1.5

1. Show that a formula ϕ is valid iff $\top \equiv \phi$, where $\top$ is an abbreviation for an instance $p \vee \neg p$ of LEM.
2. Which of these formulas are semantically equivalent to $p \rightarrow (q \vee r)$?
    (a) $q \vee (\neg p \vee r)$
    * (b) $q \wedge \neg r \rightarrow p$
    (c) $p \wedge \neg r \rightarrow q$
    * (d) $\neg q \wedge \neg r \rightarrow \neg p$.
3. An adequate set of connectives for propositional logic is a set such that for every formula of propositional logic there is an equivalent formula with only connectives from that set. For example, the set $\{\neg, \vee\}$ is adequate for propositional logic, because any occurrence of $\wedge$ and $\rightarrow$ can be removed by using the equivalences $\phi \rightarrow \psi \equiv \neg\phi \vee \psi$ and $\phi \wedge \psi \equiv \neg(\neg\phi \vee \neg\psi)$.
    (a) Show that $\{\neg, \wedge\}$, $\{\neg, \rightarrow\}$ and $\{\rightarrow, \bot\}$ are adequate sets of connectives for propositional logic. (In the latter case, we are treating $\bot$ as a nullary connective.)
    (b) Show that, if $C \subseteq \{\neg, \wedge, \vee, \rightarrow, \bot\}$ is adequate for propositional logic, then $\neg \in C$ or $\bot \in C$. (Hint: suppose C contains neither $\neg$ nor $\bot$ and consider the truth value of a formula ϕ, formed by using only the connectives in C, for a valuation in which every atom is assigned `T`.)
    (c) Is $\{\leftrightarrow, \neg\}$ adequate? Prove your answer.
4. Use soundness or completeness to show that a sequent $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ has a proof iff $\phi_1 \rightarrow \phi_2 \rightarrow \ldots \phi_n \rightarrow \psi$ is a tautology.

5. Show that the relation $\equiv$ is
   (a) reflexive: $\phi \equiv \phi$ holds for all ϕ
   (b) symmetric: $\phi \equiv \psi$ implies $\psi \equiv \phi$ and
   (c) transitive: $\phi \equiv \psi$ and $\psi \equiv \eta$ imply $\phi \equiv \eta$.
6. Show that, with respect to $\equiv$,
   (a) $\wedge$ and $\vee$ are idempotent:
      i. $\phi \wedge \phi \equiv \phi$
      ii. $\phi \vee \phi \equiv \phi$
   (b) $\wedge$ and $\vee$ are commutative:
      i. $\phi \wedge \psi \equiv \psi \wedge \phi$
      ii. $\phi \vee \psi \equiv \psi \vee \phi$
   (c) $\wedge$ and $\vee$ are associative:
      i. $\phi \wedge (\psi \wedge \eta) \equiv (\phi \wedge \psi) \wedge \eta$
      ii. $\phi \vee (\psi \vee \eta) \equiv (\phi \vee \psi) \vee \eta$
   (d) $\wedge$ and $\vee$ are absorptive:
   * i. $\phi \wedge (\phi \vee \eta) \equiv \phi$
      ii. $\phi \vee (\phi \wedge \eta) \equiv \phi$
   (e) $\wedge$ and $\vee$ are distributive:
      i. $\phi \wedge (\psi \vee \eta) \equiv (\phi \wedge \psi) \vee (\phi \wedge \eta)$
   * ii. $\phi \vee (\psi \wedge \eta) \equiv (\phi \vee \psi) \wedge (\phi \vee \eta)$
   (f) $\equiv$ allows for double negation: $\phi \equiv \neg\neg\phi$ and
   (g) $\wedge$ and $\vee$ satisfies the de Morgan rules:
      i. $\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$
   * ii. $\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$.
7. Construct a formula in CNF based on each of the following truth tables:
* (a)

p	q	ϕ_1
T	T	F
F	T	F
T	F	F
F	F	T

* (b)

p	q	r	ϕ_2
T	T	T	T
T	T	F	F
T	F	T	F
F	T	T	T
T	F	F	F
F	T	F	F
F	F	T	T
F	F	F	F

(c)

r	s	q	ϕ_3
T	T	T	F
T	T	F	T
T	F	T	F
F	T	T	F
T	F	F	T
F	T	F	F
F	F	T	F
F	F	F	T

* 8. Write a recursive function `IMPL_FREE` which requires a (parse tree of a) propositional formula as input and produces an equivalent implication-free formula as output. How many clauses does your case statement need? Recall Definition 1.27 on page 32.

* 9. Compute `CNF (NNF (IMPL_FREE` $\neg(p \to (\neg(q \wedge (\neg p \to q))))))$.

10. Use structural induction on the grammar of formulas in CNF to show that the 'otherwise' case in calls to `DISTR` applies iff both η_1 and η_2 are of type D in (1.6) on page 55.

11. Use mathematical induction on the height of ϕ to show that the call `CNF (NNF (IMPL_FREE` ϕ`))` returns, up to associativity, ϕ if the latter is already in CNF.

12. Why do the functions `CNF` and `DISTR` preserve NNF and why is this important?

13. For the call `CNF (NNF (IMPL_FREE (`ϕ`)))` on a formula ϕ of propositional logic, explain why
    (a) its output is always a formula in CNF
    (b) its output is semantically equivalent to ϕ
    (c) that call always terminates.

14. Show that all the algorithms presented in Section 1.5.2 terminate on any input meeting their precondition. Can you formalise some of your arguments? Note that algorithms might not call themselves again on formulas with smaller height. E.g. the call of `CNF (`$\phi_1 \vee \phi_2$`)` results in a call `DISTR (CNF(`ϕ_1`), CNF(`ϕ_2`))`, where `CNF(`ϕ_i`)` may have greater height than ϕ_i. Why is this not a problem?

15. Apply algorithm `HORN` from page 66 to each of these Horn formulas:
    * (a) $(p \wedge q \wedge w \to \bot) \wedge (t \to \bot) \wedge (r \to p) \wedge (\top \to r) \wedge (\top \to q) \wedge (u \to s) \wedge (\top \to u)$
    (b) $(p \wedge q \wedge w \to \bot) \wedge (t \to \bot) \wedge (r \to p) \wedge (\top \to r) \wedge (\top \to q) \wedge (r \wedge u \to w) \wedge (u \to s) \wedge (\top \to u)$
    (c) $(p \wedge q \wedge s \to p) \wedge (q \wedge r \to p) \wedge (p \wedge s \to s)$
    (d) $(p \wedge q \wedge s \to \bot) \wedge (q \wedge r \to p) \wedge (\top \to s)$
    (e) $(p_5 \to p_{11}) \wedge (p_2 \wedge p_3 \wedge p_5 \to p_{13}) \wedge (\top \to p_5) \wedge (p_5 \wedge p_{11} \to \bot)$
    (f) $(\top \to q) \wedge (\top \to s) \wedge (w \to \bot) \wedge (p \wedge q \wedge s \to \bot) \wedge (v \to s) \wedge (\top \to r) \wedge (r \to p)$